Algebra II

Kyle Binder

Spring 2020

1 Category Theory

Suggested papers:

1. Eilenberg and MacLane: *General Theory of Natural Equivalences*
2. B. Mazur: *When is one thing equal to some other thing?*

The first object we study in this course is a *category*. While quite abstract, this object allows a common language to use in mathematics to describe equivalences (e.g. natural transformations vis à vis "unnatural" transformations) and relations between mathematical structures (functors).

Definition 1. A *category* $\mathcal{C}$ consists of a class of objects $\mathrm{ob}(\mathcal{C})$ and for every pair of objects a set[1] $\mathrm{Hom}_{\mathcal{C}}(A, B)$. In the case the pair $(A, B) \neq (C, D)$ these morphisms are disjoint sets. Finally, for every triple $A, B, C \in \mathrm{ob}(\mathcal{C})$ we have a map of sets $\mathrm{Hom}(A, B) \times \mathrm{Hom}(B, C) \to \mathrm{Hom}(A, C)$ satisfying the following two properties:

1. For all $A \in \mathrm{ob}(\mathcal{C})$, there is $\mathrm{id}_A \in \mathrm{Hom}(A, A)$ such that for all $B \in \mathrm{ob}(\mathcal{C})$ and $g \in \mathrm{Hom}(A, B)$, $g = g \circ \mathrm{id}_A$ and for $f \in \mathrm{Hom}(B, A)$, $f = \mathrm{id}_A \circ f$.
2. Composition is associative.

I assume here the reader is familiar with common examples of categories, such as <u>Set</u>, <u>Mon</u> (for monoids), <u>Grp</u>, and <u>Ab</u> (for abelian groups).

N.B. The existence of morphisms between objects in a category is not necessarily guaranteed unless one looks at the morphisms of an object to itself, in which case the identity morphism always exists.

We can express statements about morphisms in terms of commutative diagrams. For example, the identity can be expressed saying the following diagram commutes:

$$\begin{tikzcd}
B \arrow[d, "g"] \arrow[dr, "g"] & \\
A \arrow[r, "\mathrm{id}_A"] \arrow[dr, "f"'] & A \arrow[d, "f"] \\
& B
\end{tikzcd}$$

and the associativity of composition can be expressed as the commutivity of

[1] It is not necessary that this is a set; it could very well be a class. With this restriction, we are technically defining a *locally small* category. When $\mathrm{ob}(\mathcal{C})$ is a proper set, we call $\mathcal{C}$ a *small category*.

$$\begin{tikzcd}
 & B \arrow[r, "g"] \arrow[dr, "h"] & C \arrow[dl, "h\circ g"] \\
A \arrow[ur, "f"] \arrow[ur, "g\circ f"'] \arrow[rr, "h\circ g\circ f"'] & & D
\end{tikzcd}$$

Subcategories

Just as with other mathematical machinary, such as sets, groups, or topological spaces, we should have a way of restricting the focus a category. We do this by way of *subcategories*.

Definition 2. A category $\mathcal{D}$ is a *subcategory* of $\mathcal{C}$, if $\mathrm{ob}(\mathcal{D})$ is a subclass of $\mathrm{ob}(\mathcal{C})$, the morphisms between objects in $\mathcal{D}$ are morphisms between the same objects in $\mathcal{C}$ (i.e. $\mathrm{Hom}_{\mathcal{D}}(A, B) \subset \mathrm{Hom}_{\mathcal{C}}(A, B)$), and the composition law in $\mathcal{D}$ is the one inherited from $\mathcal{C}$.

N.B. In class it was asked whether subcategories also have to inherit the same identity elements; it was stated that it must, but this does not seem immediate from definition

Thus, we see that subclasses are simply restrictions of a cateogry both in the objects and the morphisms; the compositional structure of the arrows are preserved. In the case the subcategory only restricts the objects and not the morphisms, we say $\mathcal{D}$ is a *full* subcategory of $\mathcal{C}$ (i.e. $\mathrm{Hom}_{\mathcal{D}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B)$ whenever the left-hand side is defined).

N.B. The compositional structure of the morphisms of an object to itself should look familiar: it is a monoid. Moreover, from any monoid M , we can create a category $\underline{M}$ with one object whose morphisms give the monoid M.

Special Morphisms

A special type of morphism in a category is an *isomorphism*. In the case of cateogries one has previously encountered in math, these are exactly the same as isomorphisms in the traditional sense.

Definition 3. We say $f \in \mathrm{Hom}(A, B)$ is an *isomorphism* if there is $g \in \mathrm{Hom}(B, A)$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.

N.B. Note that by the uniqueness of the identity for an object in a category, the above is equivalent to specifying the morphism has a left and right inverse (which *a priori* may not be the same).

Especially in the case of constructing a category from a monoid (and giving no care as to what the object of said category was), one has the feeling that the most important bits of the cateogry are the morphisms. As we shall see later on the course, this is correct. Indeed, by Yonneda, the morphisms in some sense determine the object.

Following this theme of utilizing the importance of morphisms, one can create a new category from an old by merely reversing the direction of arrows in the category. The result of this action is the opposite category (denoted $\mathcal{C}^{op}$). This category has some use when we define contravariant functors.

Definition 4. A morphism f is called *monic* if it satisfies the left-cancellation law — $f \cdot g_1 = f \cdot g_2 \Rightarrow g_1 = g_2$.

A morphism f is called *epic* if it satisfies the right-cancellation law — $g_1 \cdot f = g_2 \cdot f \Rightarrow g_1 = g_2$.

N.B. In the case of some categories (like Set), the notion of monic and epic exactly corresponds to injectivity and surjectivity. However, as we will see below, this does not always hold (as in Ring).

Here are some easy facts that are proven as soon as one writes them down.

Facts.

1. Epic and monic are preserved under composition.
2. If $g \cdot f$ is monic, so is f.
3. If $g \cdot f$ is epic, so is g.
4. If f has a left inverse, it is monic.
5. If f has a right inverse, it is epic.

We now show an example in which monic and epic indeed correspond to injective and surjective:

Proposition 1. *Fix R a ring. In R-mod, monic and epic corresponds to injective and surjective respectively.*

Proposition 2. *The fact injective and surjective imply monic and epic respectively follows by passing through the case in Set.*

Suppose that $f : A \to B$ is not injective. Then $\ker f$ is an R-mod, and we have morphisms $i : \ker f \to A$ by inclusion and $z : \ker f \to A$ by the 0-map. It is immediate to see these are different morphisms, as $\ker f \neq 0$. However, $f \cdot i = f \cdot z$, implying f is not monic.

Likewise, if $f : A \to B$ is not surjective, the cokernel $B/f(A)$ is non-zero, and the morphisms $B \to B/f(A)$ by the 0-map and projection are not equal, but their compositions with f are; thus, f is not epic.

N.B. Note that much of the above proof also applies to Grp, except epic implying monic, as $f(A)$ may not be a group. A bit more care is needed to deal with this case, but it still holds.

In the category Ring the above follows except epic may not imply surjective. The proof is the same, expect to show monic implies injective, one uses the subring of $A \times A$ consisting of pairs (a, b) where $f(a) = f(b)$ and the morphisms of projection onto the first and second coordinate, respectively. The classic example of an non-surjective epimorphism is $\mathbb{Z} \hookrightarrow \mathbb{Q}$. In general, when we have a morphism $R \to T$ where the only new data in $T \setminus R$ consists only of inverses of elements of R we have an epimorphism — the structure of the ring and homomorphism determines what these inverse get sent to.

Functors and Naturality

Definition 5. A *covariant functor* F between categories $\mathcal{C}, \mathcal{D}$ consists of a map $F : \mathrm{ob}(\mathcal{C}) \to \mathrm{ob}(\mathcal{D})$ and for every pair of $A, B \in \mathrm{ob}(\mathcal{C})$ a map $F : \hom_{\mathcal{C}}(A, B) \to \hom_{\mathcal{D}}(F(A), F(B))$ satisfying:

1. $F(f \cdot g) = F(f) \cdot F(g)$ whenever $f \cdot$ is defined.
2. $F(id_A) = id_{F(A)}$.

This definition is the same as saying

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[dr, "f\cdot g=h"'] & B \arrow[d, "f"] \\ & C \end{tikzcd}$$

implies

$$\begin{tikzcd} F(A) \arrow[r, "F(g)"] \arrow[dr, "F(f)\cdot F(g)=F(h)"'] & B \arrow[d, "F(f)"] \\ & F(C) \end{tikzcd}$$

Definition 6. A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ consists of $F : \mathrm{ob}(\mathcal{C}) \to \mathrm{ob}(\mathcal{D})$ and a map on morphisms $F : \hom_{\mathcal{C}}(A, B) \to \hom_{\mathcal{D}}(F(B), F(A))$ such that

1. $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$.
2. $F(f \cdot g) = F(g) \cdot F(f)$.

This composition law can be written as

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[dr, "f\cdot g=h"'] & B \arrow[d, "f"] \\ & C \end{tikzcd}$$

implies

$$\begin{tikzcd} F(A) & F(B) \arrow[l, "F(g)"'] \\ & F(C) \arrow[u, "F(f)"'] \arrow[ul, "F(h)=F(g)\cdot F(f)"] \end{tikzcd}$$

Example 1. The identity functor $1_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$.

Example 2. If $\mathcal{D}$ is a subcategory of $\mathcal{C}$, we have an *injection functor* $\mathcal{D} \hookrightarrow \mathcal{C}$

Example 3. A class fo functors are *forgetfull functors* which basically forget certain structures in the objects of the category.

For an example, $\underline{Mon} \to \underline{Set}$ is given by $F(M)$ is the underlying set of the monoid M. For $f \in \hom_{\underline{Mon}}(A, B)$, $F(f)$ is just viewed as f as a set map.

However, the functor on the morphisms are largely non-surjective.

Similarly we have forgetful functors
$\underline{Grp} \to \underline{Set}$
$\underline{Ring} \to \underline{Set}$
$\underline{Ring} \to \underline{Ab}$ by forgetting the multiplicative structure,
and $\underline{Ring} \to \underline{Mon}$ by forgetting the additive structure.

Example 4. We have $\underline{Grp} \to \underline{Ab}$ by $F(G) = G/(G, G)$ and $f : G \to H$ is sent to $F(f) : G/(G, G) \to H/(H, H)$ by the induced map.

Example 5. We have $\underline{Ring} \to \underline{Grp}$ by $F(R) = R^{\times} = U(R)$.

Here are some examples of *contravariant functors*:

Example 6. The *opposition* or *dual functor* $\mathcal{C} \to \mathcal{C}^{op}$ is defined by $F(A) = A$ and $F(g) = g$.

N.B. One can compose functors to get another functor.

By composing with the opposition functor, any covariant functor $\mathcal{C} \to \mathcal{D}$ can be made into a contravariant functor $\mathcal{C}^{op} \to \mathcal{D}$.

Example 7. Another duality functor is $D : R - \underline{mod} \to \underline{mod} - R$ by $M \mapsto M^* = \hom(M, R)$.

Note that M^* is naturally a right R-module — we define addition and scalar multiplication by $(f + g)(x) = f(x) + g(x)$ and $f \cdot r(x) = f(x) \cdot r$.

The functor on the morphisms acts as follows — For $L \in \hom(M, N)$, $D(L) : \hom(N, R) \to \hom(M, R)$ by $D(L)(\phi) = \phi \circ L$. Sometimes we denote $D(L) = L^*$, and it is the *transpose*.

Definition 7. A functor $F : \mathcal{C} \to \mathcal{D}$ is *faithful* if the map on morphisms is injective.

Example 8. All the forgetful functors are faithful — the morphisms are exactly determined by their actions on the underlying sets.

Definition 8. A functor is *full* if the map on morphisms is surjective.

Example 9. The duality functor $D : R - \underline{mod} \to \underline{mod} - R$ is faithful and full.

Definition 9. We say functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ are *inverses* if $G \circ F = 1_{\mathcal{C}}$ and $F \circ G = 1_{\mathcal{D}}$.

Definition 10. If $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{C} \to \mathcal{D}$ are covariant functors, a *natural transformation*[2] $\eta : F \to G$ assigns for every $A \in \mathrm{ob}(\mathcal{C})$ a morphism $\eta_A \in \hom_{\mathcal{D}}(F(A), G(A)$ such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[d, "f"] & F(A) \arrow[r, "\eta_A"] \arrow[d, "F(f)"] & G(A) \arrow[d, "G(f)"] \\ B & F(B) \arrow[r, "\eta_B"] & G(B) \end{tikzcd}$$

This means $G(f) \circ \eta_A = \eta_B \circ F(f)$ for all $f \in \hom_{\mathcal{C}}(A, B)$.

In the case η_A is an isomorphism for all $A \in \mathrm{ob}(\mathcal{C})$, η is called a *natural isomorphims*[3].

In this case, we can define an inverse natural transform $\eta^{-1} : G \to F$ by $(\eta^{-1})_A = \eta_A^{-1}$.

Easily, we can see there is an identity natural transformation for any functor.

We now want to show that there is a natural transformation from the identity functor to the double dual functor.

"The engineer exists"

Example 10. The original example from Eilenberg Mac Lane. We have the contravariant duality functor $D : R - \underline{mod} \to \underline{mod} - R$ by $M \mapsto \hom(M, R)$.

$$\begin{tikzcd} M \arrow[d, "L"] & M^* & M^{**} \arrow[d, "L^{**}"] \\ N & N^* \arrow[u, "L^*"] & N^{**} \end{tikzcd}$$

So D^2 is contravariant, $D^2(M) = M^{**}$ Recall the definition of $L^* : g \cdot L = L^*(g)$.

This means the definition of L^{**} is

[2] Also known as a "natty transform"

[3] A.K.A a "natty ice"

$$\begin{tikzcd} N^* \arrow[r, "L^*"] \arrow[dr, "L^{**}(\phi)=\phi\cdot L^*"'] & M^* \arrow[d, "\phi"] \\ & R \end{tikzcd}$$

If $g : N \to R$ (i.e. $g \in N^*$) then $(L^{**}\phi)(g) = \phi(L^*(g)) = \phi(g \cdot L)$.

If $x \in M$ then x definies an element $\eta_M(x) \in M^{**}$ by evaluation — $\eta_M(x)(f) = f(x)$ where $f \in \hom(M, R)$. The claim is that η_M defines a natural transformation $\mathrm{id}_{R-\underline{mod}} \to D^2$.

$$\begin{tikzcd} M \arrow[d, "L"] & M \arrow[r, "\eta_M"] \arrow[d, "L"] & M^{**} = D^2(M) \arrow[d, "L^{**}"] \\ N & N \arrow[r, "\eta_N"] & N^{**} = D^2(N) \end{tikzcd}$$

We have η a natural transform if this diagram commutes for all M, N, L. Let $x \in M$. $L(x) \in N$. Then $\eta_N(L(x))(g) = g(L(x)) = (gL)(x) \in R$. In the other direction if $\phi = \eta_M(x) \in M^{**}$, then

$$L^{**}(\eta_M(x))(g) = \eta_M(x)(L^*g) = (L^*g)(x) = (gL)(x)$$

Therefore the diagram commutes and D^2 is a natural transformaiton.

N.B. Note that if $R = k$ is a field, then $K - \underline{mod} = K - \underline{vect}$. If moreover we restrict to finite dimensional case, we actually get a natural isomorphism.

Also, the signle dual operation is not natural, as it requires choosing a basis.

Example 11 (Abelianization)**.** $F : \underline{Grp} \to \underline{Ab}$ by $G \mapsto G/(G, G) = G^{ab}$. We can compose this with the injection functor to get $F' : \underline{Grp} \to \underline{Ab} \to \underline{Grp}$. Letting $\nu_G : G \to G^{ab}$ we have the following commuting diagram:

$$\begin{tikzcd} G \arrow[r, "\nu_G"] \arrow[d, "f"] & G^{ab} \arrow[d, "F'(f)"] \\ H \arrow[r, "\nu_H"] & H^{ab} \end{tikzcd}$$

ν then defines a natural transformation.

Isomorphisms and Equivalences of Categories

Definition 11. We say categories $\mathcal{C}, \mathcal{D}$ are *isomorphic* if we have functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ such that $G \circ F = 1_{\mathcal{C}}$ and $F \circ G = 1_{\mathcal{D}}$.

This is a really strong equivalence, and when we do so, we identify the categories.

Example 12. $\underline{Ab}$ is isomorphic to $\mathbb{Z} - \underline{mod}$.

$R - \underline{mod}$ is isomorphic to $\underline{mod} - R^{op}$ wher R^{op} just reverses the multiplication operation.

The more interesting notion of equivalence is *equivalence of categories.*

Definition 12. $\mathcal{C}, \mathcal{D}$ are *equivalent* if $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ and $G \circ F \cong 1_{\mathcal{C}}$ and $F \circ G \cong 1_{\mathcal{D}}$ where $\cong$ indicates a natural isomorphism.

N.B.
- Isomorphic categories are equivalent.

- Equivalence of cateogries is an equivalence relation. This really says we can compose natural isomorphisms.

Given an isomorphism of categories, then F and G are honest inverses and F determines G. However, this is not true for a simple equivalence — F can vary by isomorphisms on its map of objects. Can one tell, just on level of F whether F determines an equivalence of categories? The answer is yes, as the following proposition gives us.

Proposition 3. *$F : \mathcal{C} \to \mathcal{D}$ is an equivalence of categories if and only if F is fully faithful and essentially surjective*

Proof. If F is an equivalence of categories, $G \circ F \cong 1_{\mathcal{C}}$ then $\hom_{\mathcal{C}}(A, B) \cong \hom_{\mathcal{C}}(GF(A), GF(B))$ So F must by faithful. Similarly $F \circ G \cong 1_{\mathcal{D}}$ $\hom_{\mathcal{D}}(A, B) \cong \hom_{\mathcal{D}}(FG(A), FG(B))$, so F is surjective on the morphisms and full. Finally, if $A' \in \mathrm{ob}(\mathcal{D})$ then $F \circ G \cong 1_{\mathcal{D}}$, so there is an isomorphism $\eta_{A'}$ in $\mathcal{D}$ between $FG(A')$ and A'. This gives that F is essentially surjective (its range is surjective up to isomorphisms in $\mathcal{D}$.)

We construct $G : \mathcal{D} \to \mathcal{C}$ determining an equivalence of categories with F. SKETCH: Use 3 to define $\mathcal{D}$ on objects — for $A' \in \mathrm{ob}(\mathcal{D})$ take $A \in \mathrm{ob}(\mathcal{C})$ with $F(A)$ isomorphic to A'. Define $G(A') = A$. On morphisms, we have F fully faithful, so we can uniquely find inverses. $\square$

$$\begin{tikzcd}
A' \arrow[r, "\eta_{A;}"] \arrow[d, "f'"] & F(A) \arrow[d, dotted, "F(f)"] \\
B' \arrow[r, "\eta_{B'}"] & F(B)
\end{tikzcd}$$

$\eta_{B'} \circ f' \circ \eta_{A'}^{-1} \in \hom(F(A), F(B))$. Then use fully faithful to say $\exists! f \in \hom(A, B)$ so that the above diagram commutes. Take $G(f') = f$.

One then must check G defines a functor and (F, G) give equivalences.

Example 13. Let R be a ring, $M_n(R)$ $n \times n$ matrices over R. Then $\underline{mod} - R$ and $\underline{mod} - M_n(R)$ are equivalent:

Proof. $F : \underline{mod} - R \to \underline{mod} - M_n(x)$ by M R-module, let $M^{(n)} = M \oplus \cdots \oplus M$. Say $F(M) = M^{(n)}$ and for $f \in \hom(M, N)$ let $F(f) = f^{(n)}$ where $f^{(n)}(m_1, \ldots, m_n) = (f(m_1), \ldots, f(m_n))$. Check this is fully faithful and essentially surjective. $\square$